Axiom: AI for Differentiated Geometry Learning

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Core concept. Axiom teaches the Pythagorean theorem through one mathematical relationship, $a^2 + b^2 = c^2$. Its responsive web interface uses an X-inspired feed, original triangle illustrations, three difficulty controls, progressive hints, immediate feedback, and a worked solution.

Differentiation. Foundation asks for the hypotenuse with both legs given (6 cm and 8 cm → 10 cm) and scaffolds identification, substitution, and square roots. Standard embeds a missing-leg problem in a ladder scenario (13 m ladder, 5 m from the wall → 12 m high), requiring modeling and subtraction. Advanced asks for the shortest surface path between opposite corners of a 3 × 4 × 12 cm box. Learners unfold adjacent faces and compare √193, √241, and √265, giving 13.89 cm. The space diagonal is explicitly excluded. Across levels, numerical variants, context, conceptual demand, and guidance change while the core theorem stays fixed.

AI architecture. An open-source SmolLM2-135M-Instruct model runs through Transformers.js/ONNX in a browser Web Worker using quantized q4 weights. It loads on demand; model files are cached where the browser permits. Inference happens on the learner's device, so no API key or remote inference backend is necessary. AI explains a verified hint, while separate JavaScript mathematics controls grading and solutions. A download failure is shown explicitly; deterministic guidance is not represented as generated AI.

System prompts. The shared instruction is: “You are Axiom, a concise, encouraging geometry tutor. Explain only the supplied verified hint. Do not change numbers or invent calculations. Do not give the final answer. Use at most three short sentences in plain English.” Foundation adds “Use very simple language and one small step at a time.” Standard adds “Explain the relation between the real-world situation and a right triangle.” Advanced adds “Encourage spatial reasoning. Explain unfolding faces and comparing all three routes.” The user message supplies the current level, problem, and verified hint. Generation uses deterministic decoding, a repetition penalty, and a 100-token limit.

Feedback and validation. Three progressively specific hints are available at every level. Correctness tests cover 126 generated cases, all three box unfoldings, known answers, answer parsing, and rounding; five automated test groups pass. Browser checks exercise difficulty switching, answer feedback, hints, solutions, variations, persistent progress, bookmarks, responsive navigation, and model-download error handling. Viewing the worked solution prevents independent-completion credit for that attempt. AI text remains advisory and may contain mistakes.

Delivery and limitations. Source, README, this report, and a GitHub Pages deployment workflow are included. Production build and local learner flows are validated. The cloud proxy blocks Hugging Face downloads and GitHub API access; successful real-model inference and a published live URL remain unverified until access and deployment are completed. Progress is local to one browser. The implementation was prepared with AI coding assistance; mathematical results were checked independently of the language model.